MOSAIC

Reference-Grounded Compositional Synthesis for Multi-Source Emoji Fusion

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Identity · Semantics · Geometry

1 One object, several source identities

Fusion requires a carrier, transferable traits and compatible attachment regions. We expose these decisions as reusable conditions for generation.

I Appearance

RGB pair → U-Net / SEUNet

L1 + SSIM; EMA-stabilized reconstruction

II Composition

separation → carrier → scale / offset

position-aware source contributions

III Semantics

descriptions → LLM → FLUX.1 + LoRA

open relations; text compresses appearance

2 COMPOSITION AS AN EXPLICIT LATENT INTERFACE

Proposed unified architecture / diagrams specify mechanisms, not measured outputs

A Source-resolved representation





EmojiSource

identity grapheme + codepoints
meaning semantic description
reference canonical image

source-specific visual encoding

F_1: contour / palette

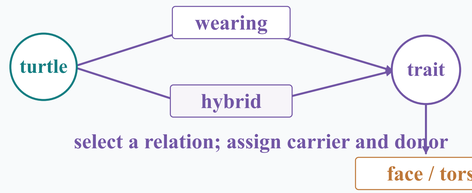
F_2: local traits

concept descriptions

turtle: shell, green body;
scientist: goggles, coat, professional persona

visual references $V = \{v_1, \ldots, v_n\}$

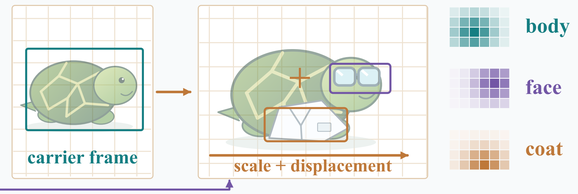
B Relational composition planner



P.role	carrier = turtle; donor = scientist
P.transfer	goggles → eyes; lab coat → torso
P.constraint	retain shell; exclude independent human

structured semantic plan P

C Geometry and region allocation

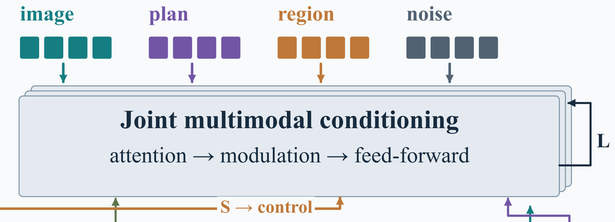


$T_i(u) = s_i\, u + \Delta_i;\ \Delta_i = (\Delta x_i, \Delta y_i)$

S = {role weights, transforms, region masks}

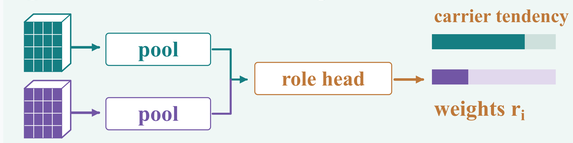
Region-specific influence ties an attribute to an attachment site rather than the entire image.

D Reference-conditioned synthesis



h(t) → velocity prediction → h(t − Δt) → decode

a Carrier selection and spatial reuse

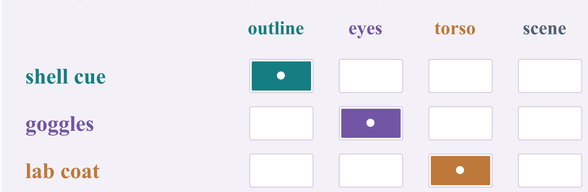


center detector → scale / offset proposal

separate F_i → transform $T_i(F_i)$ → local mixing

EmojiSeparator, CenterAwareScaleModule and PositionalAttention motivate a reusable structural prior.

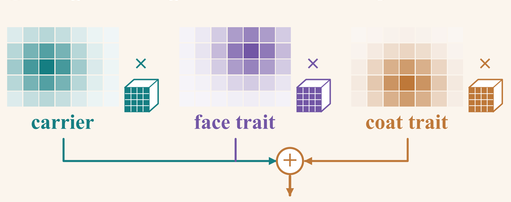
b Typed trait-to-region correspondence



preserve identity + transfer selected attributes

Bind each trait to its donor and attachment region while preserving the carrier's visual identity.

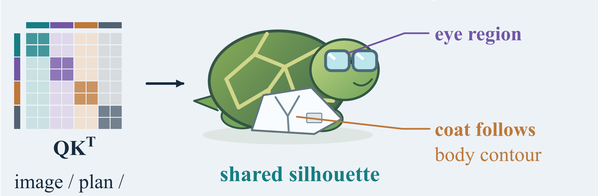
c Region-weighted feature composition



$F(x, y) = \Sigma_i\, a_i(x, y) \cdot T_i(F_i)(x, y)$

Local contributions: $a_i \geq 0$, $\Sigma_i\, a_i = 1$. Export masks or control hints through supported renderer interfaces.

d Joint attention with a shared output



Factorized design: $p(Y, P, S \mid E) = p(P \mid E) \cdot p(S \mid E, P) \cdot p(Y \mid V, P, S; \theta, \Delta\theta_{style})$

role → relation → attachment → appearance

3 Cross-generation optimization

paired fusion images → **structure teacher / geometric prior**

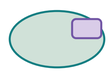
source descriptions + relations → **FusionPlan supervision**

emoji-domain image corpus → **renderer style adaptation**

Staged integration preserves the pixel model's structural knowledge while separating semantic planning from domain appearance.

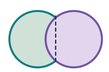
4 Compositional modes share one interface

Carrier–attribute



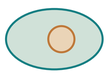
asymmetric roles

Balanced hybrid



shared identity

Embedded / accessory



local contribution

The same source set admits different relations. P selects the compositional mode; S locates its visual realization; V retains the original appearance.

5 Mechanism-oriented evaluation design

Condition removed	Mechanism to inspect	Failure probe
visual references	source fidelity	identity drift
structured plan	relation selection	co-occurrence
spatial prior	trait localization	misattachment
style adapter	domain rendering	scene / style drift

Hold inputs and sampling settings fixed; inspect each factor independently.

MOSAIC: Multi-source cOmposition with Semantic And Identity Constraints

Historical components inform the proposed integration; all visual examples are schematic.

Design context: [1] UNO, arXiv:2504.02160 (2025) — multi-subject conditioning. [2] OmniGen2, arXiv:2506.18871 (2025) — decoupled semantic / visual pathways.

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